Exam-style | No answers | File 6 / 7

Q Chapter 2 exam-style questions

أسئلة امتحانية حقيقية: تصنيف مع تبرير، بناء وقراءة، اكتشاف أخطاء. ليست أسئلة مباشرة

Built from the outline's assessment criteria for this chapter. Every question carries its lesson tag and outcome tag; a wrong answer points to the lesson file to reopen. Answers are in a **separate file**.

LESSON	2-1	2-2	2-3
Questions touching the lesson	16	15	15

طريقة الاستخدام

الأسئلة على نمط الكويز والميدتيرم: **تصنيف مع تبرير**، **بناء جدول أو رسم**، **قراءة رسم واستخراج قيمة**، و**اكتشاف خطأ طالب**.
حلّها مغلق الكتاب ثم صحّح من ملف الإجابات. الجزء F محاكاة لجزء هذا الفصل في الكويز الصفي الأول.

Part A · Concept and interpretation (multiple choice)

Give the letter and a one-line reason.

A1 2-1 CLO2-B

A grouped distribution of whole-number ages has classes 35–44, 45–54, 55–64. The class width is:

(a) 9 (b) 9.5
(c) 10 (d) 11

A2 2-1 CLO2-B

Data range from 18 to 60 and 6 classes are required. The class width should be:

(a) 7 (b) 7.5
(c) 8 (d) 6

A3 2-1 CLO2-B

Which is **not** a requirement of a grouped frequency distribution?

(a) classes are mutually exclusive (b) classes are continuous
(c) all classes have the same width (d) every class contains at least one value

A4 2-1 CLO2-B

A class has relative frequency 0.15 in a data set of 60 values. Its frequency is:

(a) 6 (b) 9

(c) 15 (d) 0.25

A5 2-2 CLO3-A

On a frequency polygon, the points are plotted against the:

(a) class limits (b) lower class boundaries

(c) class midpoints (d) upper class boundaries

A6 2-2 CLO3-A

The first point of an ogive is plotted at:

(a) (first midpoint, first frequency) (b) (first lower boundary, 0)

(c) (first upper boundary, 0) (d) (0, 0)

A7 2-2 CLO3-B

Part of an ogive is perfectly horizontal. This means:

(a) that class contains the most values (b) that class has frequency 0

(c) the data are uniform (d) the graph is drawn incorrectly

A8 2-2 CLO3-B

A histogram's tallest bar is near the left and the bars trail off to the right. The distribution is:

(a) left-skewed (b) right-skewed

(c) J-shaped (d) uniform

A9 2-3 CLO2-C

The nationality of 200 tourists is best displayed with a:

(a) histogram (b) ogive

(c) bar graph (d) time series graph

A10 2-3 CLO2-C

In a pie chart, a category holding 15% of the data has a sector of:

(a) 15° (b) 36°

(c) 54° (d) 150°

A11 2-3 CLO3-B

In a stacked bar, the "female" section runs from 20 up to 35 on the y-axis. The number of females is:

(a) 35 (b) 20

(c) 15 (d) 55

A12 2-3 CLO3-A

A stem-and-leaf plot's key reads 4 | 7 = 0.47. The row "3 | 2 8" represents:

(a) 3.2 and 3.8 (b) 32 and 38

(c) 0.32 and 0.38 (d) 0.032 and 0.038

Part B · Short exam problems by lesson

Construction, completing tables, reading graphs, choosing graphs, find-the-mistake.

B1 2-1 CLO2-B

Waiting times (minutes) of 30 patients: 12, 25, 31, 18, 44, 27, 35, 22, 15, 29, 38, 41, 20, 33, 26, 19, 47, 30, 24, 36, 28, 16, 39, 21, 32, 45, 23, 34, 27, 30. Construct a grouped frequency distribution with **5 classes**, including boundaries, f, RF and CF.

B2 2-1 CLO2-B

Complete the table (n = 40):

CLASS	F	RF	CF
10–19	3		3
20–29	?		11
30–39	12		?
40–49	?		36
50–59	?		40

Then find the percentage of values below 39.5.

B3 2-1 CLO2-B

A distribution has class boundaries 15.5–20.5, 20.5–25.5, 25.5–30.5, 30.5–35.5. Find the class limits, the class width and the midpoints.

B4 2-1 2-3 CLO2-B · CLO3-B

The majors of 40 College of Science students: Statistics 9, Mathematics 12, Physics 6, Chemistry 8, Biology 5. Build the categorical distribution with percentages and pie angles, and write one sentence of interpretation.

B5 2-2 CLO3-B

Use the cumulative frequencies of the record-temperature ogive (Example 2-4): less than 99.5 → 0, 104.5 → 2, 109.5 → 10, 114.5 → 28, 119.5 → 41, 124.5 → 48, 129.5 → 49, 134.5 → 50. (a) How many states had a record of at least 114.5 °F? (b) What percentage had records between 109.5 and 124.5 °F? (c) Which class has the most states, and how can you tell from the ogive?

B6 2-2 CLO2-C

For the classes 20–29 (f 4), 30–39 (f 9), 40–49 (f 15), 50–59 (f 7), 60–69 (f 5), list every point you would plot for (a) the frequency polygon and (b) the ogive.

B7 2-2 CLO3-B

Name the shape of each histogram, read left to right: (a) 2, 5, 9, 14, 10, 6, 3 (b) 15, 10, 6, 4, 2, 1 (c) 8, 3, 2, 3, 9 (d) 3, 9, 4, 3, 8, 3 (e) 6, 6, 7, 6, 6, 7

B8 2-2 CLO2-C · CLO3-A

Find the mistakes. A student drew a histogram of the temperature data with bars centred on 100–104, 105–109, ..., a gap between bars, no axis labels, and the y-axis starting at 2.

B9 2-3 CLO2-C

40 students travel to SQU by bus 14, car 18, walk 5, taxi 3. Find each pie angle and percentage. Which two categories together make up more than three-quarters of the circle?

B10 2-3 CLO3-B

A stacked bar graph of Year 1 to Year 3 STAT 2101 enrolments has male sections from 0 to 14, 0 to 18 and 0 to 22, and female sections on top reaching 25, 34 and 40. Build the frequency table and state one insight.

B11 2-3 CLO2-C · CLO3-B

Draw a stem-and-leaf plot with a key for the reaction times (s): 1.3, 1.5, 1.8, 2.0, 2.2, 2.2, 2.4, 2.7, 2.9, 3.1, 3.3, 3.4, 3.4, 3.6, 3.8, 4.1, 4.5. Then give the shape and the peak.

B12 2-3 CLO3-B

Monthly sales (OMR thousands) of a Muscat shop: Jan 42, Feb 45, Mar 51, Apr 49, May 58, Jun 66, Jul 71. Describe the trend, and say between which two consecutive months the change was fastest.

B13 2-2 2-3 CLO2-C

Choose the most suitable graph and justify it: (a) the daily number of COVID cases in Oman over a year (b) favourite mobile brand of 150 students (c) the ages of 500 patients, to see the shape (d) the percentage of the national budget spent on each sector (e) how many of 500 patients are younger than 40

Part C · Long structured questions

Full table-to-graph workups. Marks in brackets.

C1 2-1 2-2 CLO2-B · CLO2-C · CLO3-B **[16 marks]**

Maximum temperatures (°C) in Muscat on 40 days: 29, 33, 35, 31, 38, 41, 36, 34, 30, 37, 39, 42, 35, 33, 36, 40, 44, 38, 32, 37, 35, 34, 39, 36, 41, 37, 33, 38, 36, 35, 27, 40, 34, 37, 36, 38, 43, 35, 39, 36.

1. Construct a grouped distribution with **6 classes**: limits, boundaries, midpoints, f, RF, CF, CRF.
2. Which class is modal? Describe the shape.
3. On how many days was the maximum below 35.5? What percentage is that?
4. List the x-values you would use for the histogram, the frequency polygon and the ogive.

C2 2-1 2-2 CLO2-B · CLO3-B **[10 marks]**

Read this distribution of exam times (minutes):

BOUNDARIES	F
29.5–39.5	4
39.5–49.5	7
49.5–59.5	13
59.5–69.5	10
69.5–79.5	6

1. Find the class width, the class limits and n.
2. Find the midpoints and the relative frequencies.
3. What percentage of students took less than 59.5 minutes?
4. Between which two boundaries does the ogive rise most steeply? Why?

C3 2-1 2-3 CLO2-B · CLO2-C · CLO3-C **[12 marks]**

60 SQU students named their preferred study place: library 21, home 18, café 9, lab 12. Among them, males chose library 9, home 11, café 4, lab 6.

1. Build the frequency distribution with percentages.
2. Find the pie-chart angles.
3. Build the male/female table and describe how a grouped and a stacked bar graph would display it.
4. Give two insights.

Part D · True / false with justification

The justification is the mark.

D1 2-1 CLO2-B · CLO3-A

True or false? Justify. A class with frequency 0 may be left out of a grouped distribution.

D2 2-1 CLO2-B · CLO3-A

True or false? Justify. If range ÷ number of classes is exactly 6, the class width is 6.

D3 2-2 CLO2-B · CLO3-A

True or false? Justify. An ogive can go down.

D4 2-2 CLO2-B · CLO3-A

True or false? Justify. A histogram and a bar graph differ only in colour.

D5 2-3 CLO2-B · CLO3-A

True or false? Justify. A stem-and-leaf plot keeps the original data values.

D6 2-3 CLO2-B · CLO3-A

True or false? Justify. The angles in a correctly drawn pie chart must add to 360°.

D7 2-2 CLO2-B · CLO3-A

True or false? Justify. A relative frequency histogram has a different shape from the frequency histogram of the same data.

D8 2-1 CLO2-B · CLO3-A

True or false? Justify. The class width can be found as (upper boundary − lower boundary) + 1.

Part F · In-Class Quiz I simulation, Chapter 2 part (15 marks · 15 minutes)

Closed book, calculator allowed.

F1 2-1 CLO2-B

[3 marks] Data range from 47 to 92 and 5 classes are required. Find the width and the first two classes (limits and boundaries).

F2 2-1 CLO2-B

[3 marks] Frequencies 6, 11, 15, 8 (n = 40). Find the CF and CRF of the third class, and the RF of the fourth class.

F3 2-2 CLO3-A

[3 marks] For the class 120–129, give the value used on the x-axis of a histogram, a frequency polygon and an ogive.

F4 2-3 CLO2-C

[3 marks] A category has 27 of 120 observations. Find its percentage and its pie angle.

F5 2-3 CLO3-B

[3 marks] Stem-and-leaf (key 5 | 2 = 52): 4 | 1 3 · 5 | 0 2 4 7 · 6 | 1 3 3 5 6 8 9 · 7 | 0 2 4 5 · 8 | 1. How many values? What is the peak? What is the shape?

Chapter 2 study set: 7 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's slide pack, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief · 01 · Lesson 2-1: Organizing Data · 02 · Lesson 2-2: Histograms, Frequency Polygons, Ogives · 03 · Lesson 2-3: Other Types of Graphs · 04 · Revision pack · **05 · Exam-style questions** · 06 · Full worked answers